

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Homework 1
  - Due **this Friday** 10/4 @5 pm
- Midterm Question Contribution
  - Google Form: https://forms.gle/PQWnzJ3EphU2WyPw8 (see the form for the detailed instruction)
  - Due next Thursday 10/10 @5 pm
  - **3 bonus points** on the midterm if you participate and additional **2 bonus points** if your question is selected!

This Lecture

- Decision Networks
  - Computing MEU of DN

Last Lecture

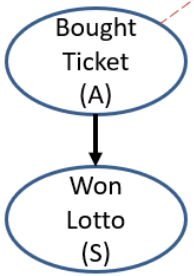
- Utility Theory: Probability -> Decision
  - **Utility Function** $U(s, a) = u$
    - mapping “state” and “action” to a “utility score”
  - **Expected Utility** $EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$
  - **Maximum Expected Utility**
    - $MEU(e) = \max_{a \in A} EU(a|e)$
    - $a^* = \arg\max_{a \in A} EU(a|e)$

Last Lecture

- Decision Networks
  - Bayesian Networks + Utility Theory
  - Nodes
    - **Chance** Nodes ○: BN variables w/ CPTs
    - **Decision** Nodes ▢: choices/actions the agent has to make
    - **Utility** Nodes ◇: utility function

Decision Networks

As Bayesian Network (BN)

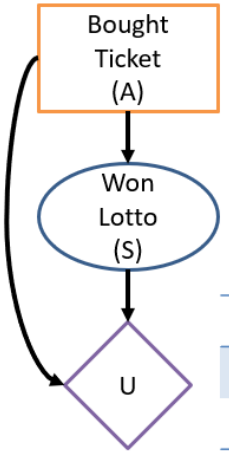


What would $P(A)$ look like?

Full Joint: $P(A,S)$

$P(S|a,e)$ query: $P(S|a) = \frac{P(S,a)}{P(a)}$

As Decision Network (DN)



Full Joint: $P(S|do(A))$

$P(S|a,e)$ query: $P(S|do(a))$

A	S	$P(S \mid do(A))$
0	0	1.0000000
1	0	0.9999999

$U(pa) = U(A,S)$ = (same as example above)

Decision Nodes

- Have **no parent**
- Have **no CPT** (assumed "given")
- Do **not** need to be **normalized**
- A.k.a. "intervention"

Utility Nodes

- **Parents** of utility nodes (pa) are the arguments to the utility function.
- **Parents** of utility nodes can be **chance** nodes OR **decision** nodes.

Expected Utility for DN

$$EU(a|e) = \sum_{s \in S} P(s|a,e) \cdot U(s,a)$$

- How to compute EU for DN?

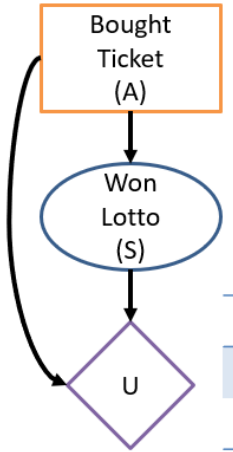
a: full assignments to decision node

$$EU(a|e) = \sum_{s \in S} P(s|do(a),e) \cdot U(pa)$$

e: **observations** about the chance nodes

s: "state" or chance node **parents** of utility nodes

As Decision Network (DN)



Full Joint: $P(S|do(A))$

$P(S|a,e)$ query: $P(S|do(a))$

A	S	$P(S \mid do(A))$
0	0	1.0000000
1	0	0.9999999

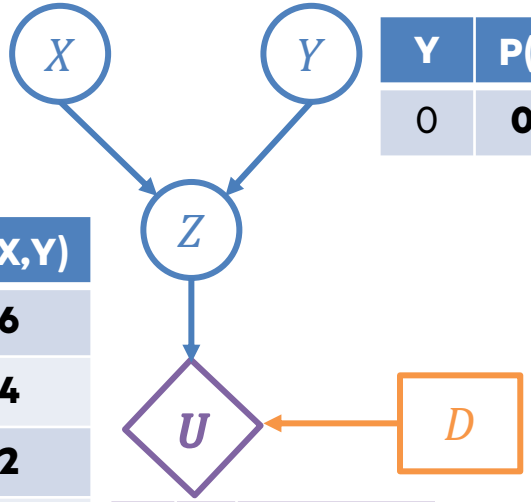
$U(pa) = U(A,S) =$ (same as example above)

Example

- Chance nodes: $X, Y, Z \in \{0,1\}$
- Decision node: $D \in \{0,1\}$
- Utility node: U
- **Problem: Find the MEU(X=0)**
  - How should we get started?

$$EU(a|e) = \sum_{s \in S} P(s|a,e) \cdot U(s,a)$$

Idea: find EU of each decision and choose the best



X	P(X)
0	**0.2**

Y	P(Y)
0	**0.3**

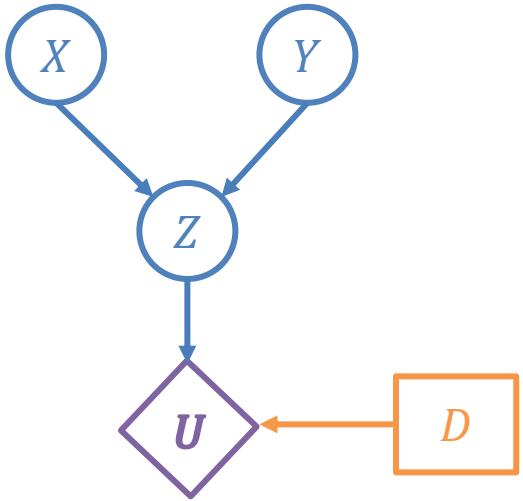
X	Y	Z	P(Z|X,Y)
0	0	0	**0.6**
0	1	0	**0.4**
1	0	0	**0.2**
1	1	0	**0.3**

Z	D	U(Z,D)
0	0	**100**
0	1	**40**
1	0	**60**
1	1	**80**

$$EU(a|e) = \sum_{s \in S} P(s|a,e) \cdot U(s,a)$$

Step 1: Compute $P(S|A, e)$

- **Step 1.1: Identify Variables**
  - Query Variables $Q = \{Z\}$
  - Evidence Variables $e = \{X = 0\}$
  - Hidden Variables $\mathrm{Y} = \{Y\}$
  - Decision Variables $D = \{D\}$
  - $P(S|A, e) = P(Q|D, e)$



Review: Compute $P(Q|e)$ in Enumeration Inference

$$P(Q|e) = \frac{P(Q,e)}{P(e)}$$

Step 4: Normalization

$$= \frac{1}{P(e)} \cdot P(Q,e)$$

$$= \frac{1}{P(e)} \cdot \sum_{y \in Y} P(Q,e,y)$$

joint distribution: can be factored w/ MF

$$= \frac{1}{\sum_{q \in Q} P(Q,e)} \cdot \sum_{y \in Y} P(Q,e,y)$$

Step 2: P(Q,e)

Step 3: P(e)

$$EU(a|e) = \sum_{s \in S} P(s|a,e) \cdot U(s,a)$$

Step 1: Compute $P(S|A,e)$

- **Step 1.2: Find $P(Q,e|D) = \sum_y P(Q,e,y|D)$**

Law of Total Prob.

$$P(Z,X=0|D) = \sum_{y \in Y} P(Z,X=0,Y=y|D)$$

Markovian Factorization

$$= \sum_{y \in Y} P(Z|X=0,Y=y,D)P(X=0|D)P(Y=y|D)$$

Can this be computed?

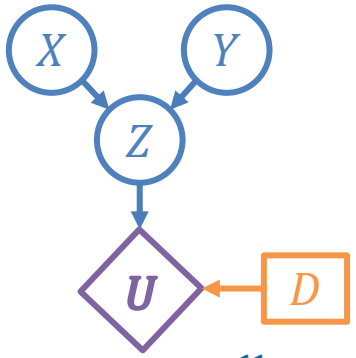
$$= \sum_{y \in Y} P(Z|X=0,Y=y)P(X=0)P(Y=y)$$

Optimize Summation Order

$$= P(X=0)\sum_{y \in Y} P(Z|X=0,Y=y)P(Y=y)$$

$$= P(X=0)(P(Z|X=0,Y=0)P(Y=0) + P(Z|X=0,Y=1)P(Y=1))$$

  - Z=0: $P(Z,X=0|D)$ =0.2*(0.6*0.3+0.4*0.7)=**0.092**
  - Z=1: $P(Z,X=0|D)$ = 0.2*(0.4*0.3+0.6*0.7)=**0.108**



$$EU(a|e) = \sum_{s \in S} P(s|a,e) \cdot U(s,a)$$

Step 1: Compute $P(S|A,e)$

- **Step 1.3: Find $P(e|D) = \sum_q P(Q = q, e|D)$**

$$P(e|D) = P(X = 0|D)$$

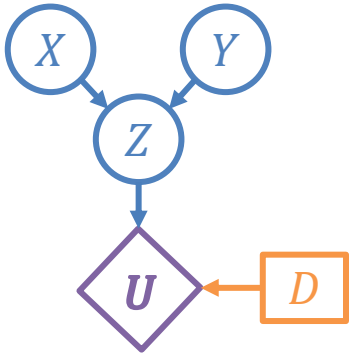
Why?

$$= P(X = 0)$$

$$= \sum_z P(Z = z, X = 0)$$

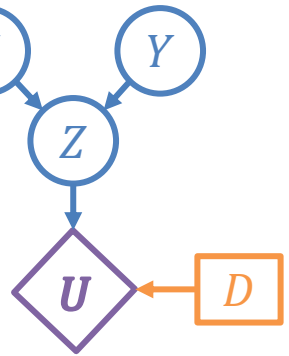
$$= P(Z = 0, X = 0) + P(Z = 1, X = 0)$$

$$= 0.092 + 0.108 = 0.2$$



Step 1: Compute $P(S|A, e)$

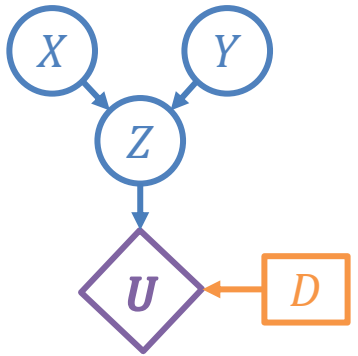
$$EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$$



- **Step 1.4: Find $P(Q|D, e) = \frac{P(Q, e|D)}{P(e|D)}$**
  - Step 1.2:
    - Z=0: $P(Z, X = 0|D)$ =0.2*(0.6*0.3+0.4*0.7)=**0.092**
    - Z=1: $P(Z, X = 0|D)$ = 0.2*(0.4*0.3+0.6*0.7)=**0.108**
  - Step 1.3: $P(e|D) = P(X = 0|D) = 0.2$
  - Step 1.4:
    - Z=0: $P(Z = 0\ |D, X = 0) = \frac{P(Z=0, X=0)}{P(X=0)}$ = 0.092/0.2 =0.46
    - Z=1: $P(Z = 1|D, X = 0) = \frac{P(Z=1, X=0)}{P(X=0)}$ = 0.108/0.2=0.54

$$EU(a|e) = \sum_{s \in S} P(s|a,e) \cdot U(s,a)$$

Step 2: Find $EU(a|e) \forall a \in A$



- $EU(a|e) = EU(D|X=0)$

  $= \sum_z P(Z=z|D,X=0) \cdot U(Z,D)$

  - D=0: $EU(a|e) = EU(D=0|X=0)$

    $= P(Z=0|X=0) \cdot U(Z=0,D=0) +$
    $P(Z=1|X=0) \cdot U(Z=1,D=0)$

    $= 0.46 * 100 + 0.54 * 60 = 78.4$

  - D=1: $EU(a|e) = EU(D=1|X=0)$

    $= P(Z=0|X=0) \cdot U(Z=0,D=1) +$
    $P(Z=1|X=0) \cdot U(Z=1,D=1)$

    $= 0.46 * 40 + 0.54 * 80 = 61.6$

Z	D	U(Z,D)
0	0	**100**
0	1	**40**
1	0	**60**
1	1	**80**

Step 3: Find $MEU(e)$ and $d^* = \text{argmax}_d EU(a|e)$

- $MEU(X = 0) = \max EU(d|X = 0)$
  $= \max(78.4, 61.6)$
  $= 78.4$
- $d^* = \arg\max MEU(X = 0) = 0$